Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	16
interior lacuna count	16
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1420.45	1293	702.329	px^2
major_axis_length_px	lacuna shape	64.3118	54.0186	22.3213	px
minor_axis_length_px	lacuna shape	35.2082	30.951	7.6188	px
aspect_ratio	lacuna shape	1.8719	1.7017	0.7153	unitless
eccentricity	lacuna shape	0.7783	0.8091	0.1414	unitless
solidity	lacuna shape	0.7237	0.7085	0.0904	unitless
roots_count	headline	8.5625	9	4.4567	count
ring_length_r30_px	headline	319.188	316.5	170.941	px
ring_length_r60_px	reported	585.062	618.5	350.038	px
owned_length_px	ownership-dependent	930.929	738.884	1052.16	px
edge_count	ownership-dependent	51.375	34.5	60.4206	count
mean_edge_length_px	ownership-dependent	18.6297	18.674	2.9212	px
perimeter_px	normalised	219.641	184.48	100.453	px
ring_area_r30_px2	normalised	6570.25	7072	2510.45	px^2
ring_area_r60_px2	normalised	15758	18338.5	6991.56	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0469825	0.0468823	0.0133093	px^-1
ring_density_r60	normalised	0.0361238	0.0342294	0.011688	px^-1
roots_per_100px_perimeter	normalised	3.9152	3.896	1.481	per 100 px
ring_attached_length_r30_px	network v2	287.688	271	173.468	px
ring_attached_length_r60_px	network v2	487.75	511	339.005	px
ring_length_w_r30_px	network v2	360.603	361.149	192.182	px
ring_length_w_r60_px	network v2	657.206	697.393	391.586	px
sholl_crossings_r10	network v2	7.6875	7.5	3.7008	count
sholl_crossings_r20	network v2	8.0625	8	4.3584	count
sholl_crossings_r30	network v2	7.75	7	4.7958	count
loop_count	loops, report only	3.125	0	6.0429	count
loop_edge_share	loops, report only	0.1207	0	0.1634	share
ring_width_mean_r30_px	width	7.2987	7.4421	1.3068	px

Per field
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measure	value	units
analysed_area_px2	869119	px^2
total_skeleton_length_px	19856	px
canalicular_length_density_per_px	0.0228461	px^-1
junction_count	428	count
junction_density_per_px2	0.00049245	px^-2
lacuna_count	16	count

lacunae_per_px2	1.841e-05	px^-2
skeleton_component_count	293	count
mean_component_length_px	67.7679	px
median_component_length_px	16	px
field_length_density_w_per_px	0.0254924	px^-1
field_density_without_flagged_per_px	0.0173825	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	563	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0228461	px^-1
field_length_density_with_unattached_per_px	0.0234939	px^-1
excluded_area_px2	161365	px^2
analysed_field_area_px2	887211	px^2
loop_count	56	count
loop_edge_share	0.1739	share
edge_count	1294	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4789	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	16813	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
2	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
3	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
4	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
5	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
6	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
7	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
8	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
9	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
10	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
11	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
12								290.753	563.455	False
13								291.903	675.169	False
14								565.978	720.266	False
15								614.615	620.067	False
16								698.601	352.387	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	380	734	854.553	50	17.0911	8.3983

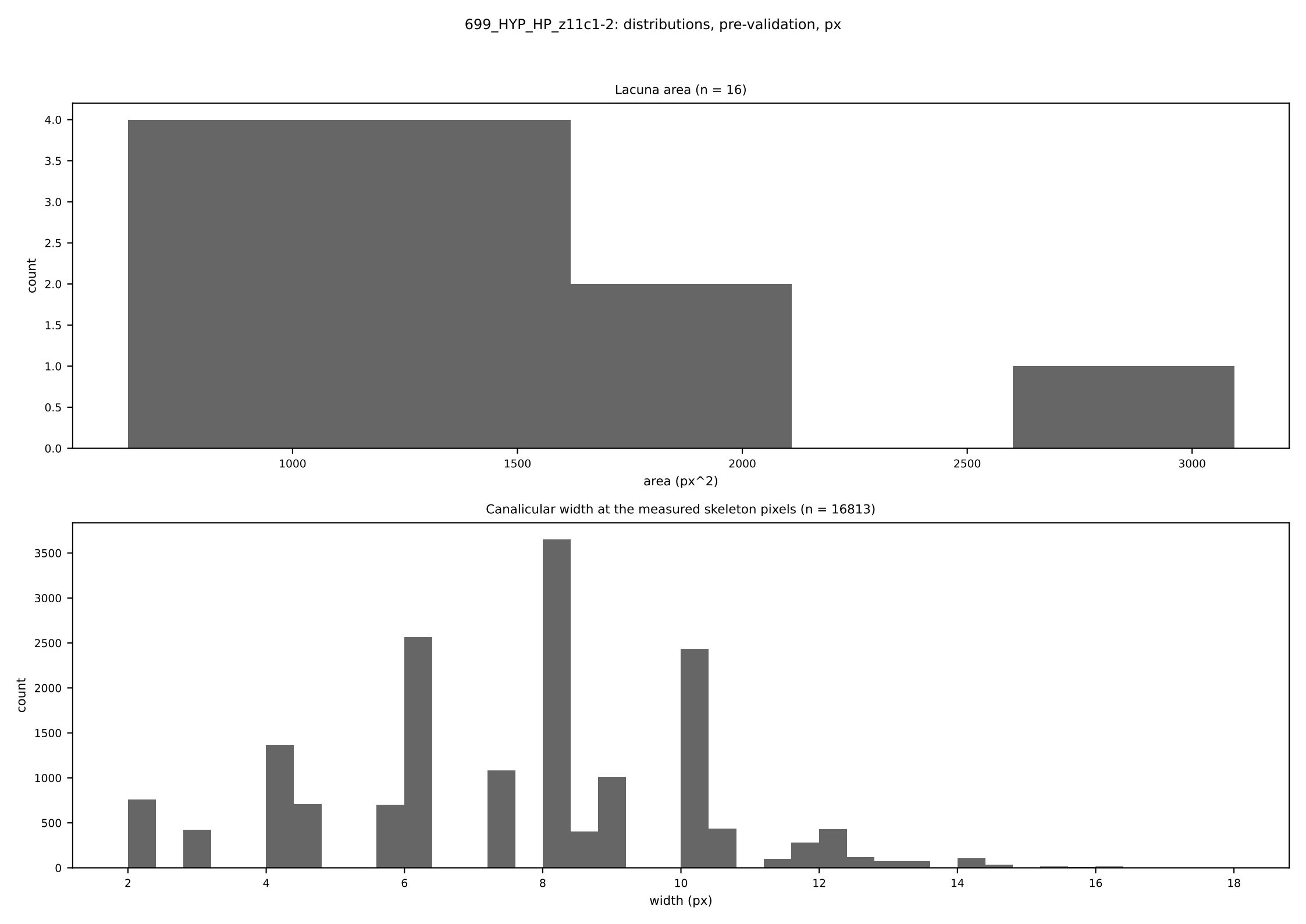
2	12	386	694	799.855	34	23.5251	8.086
3	8	302	657	1005.88	59	17.0488	9.3181
4	3	223	494	679.399	30	22.6466	7.4308
5	18	814	1543	4371.76	252	17.3482	7.4254
6	10	331	436	403.635	20	20.1817	7.7794
7	3	156	273	279.439	14	19.9599	5.4602
8	14	441	747	737.399	35	21.0685	7.4535
9	10	204	283	338.392	19	17.8101	6.2316
10	6	220	308	276.35	15	18.4234	7.1012
11	9	372	637	806.441	40	20.161	8.5806
12	11	404	860	1028.41	64	16.0689	8.2415
13	3	243	600	740.37	50	14.8074	6.5639
14	2	32	32	125.154	6	20.8591	4.1477
15	9	222	241	157.953	13	12.1503	6.3158
16	7	377	822	2289.87	121	18.9246	8.2448

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.622	7458	18845	1	1	0.050952	0.0389493	6.2623
2	234.25	8068	19848	1	1	0.0478433	0.0349657	5.1227
3	172.995	7102	19616	1	1	0.0425232	0.0334931	4.6244
4	174.066	7351	20270	1	1	0.030336	0.024371	1.7235
5	467.446	11796	28237	1	1	0.0690064	0.0546446	3.8507
6	318.333	7208	13815	1	1	0.0459212	0.0315599	3.1414
7	232.285	5734	14497	1	1	0.0272061	0.0188315	1.2915
8	409.931	10384	22845	1	1	0.0424692	0.0326986	3.4152
9	174.853	4702	8707	1	1	0.0433858	0.0325026	5.7191
10	172.267	3574	7974	1	1	0.0615557	0.0386255	3.483
11	182.894	6958	19375	1	1	0.0534636	0.0328774	4.9209
12	186.066	7155	17832	1	1	0.056464	0.0482279	5.9119
13	91.9828	5369	14348	1	1	0.0452598	0.0418177	3.2615
14	104.468	1694	2277	1	1	0.0188902	0.0140536	1.9145
15	228.35	3529	4140	1	1	0.0629073	0.0582126	3.9413
16	172.439	7042	19502	1	1	0.0535359	0.0421495	4.0594

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	351	596	434.747	830.796	8	8	13
2	374	618	434.049	765.701	9	12	12
3	272	599	349.907	743.428	6	9	7
4	185	309	248.995	557.487	7	4	7
5	800	1490	916.955	1722.34	18	18	17
6	270	337	372.392	493.475	9	9	6
7	118	168	170.468	296.894	4	5	4
8	434	530	489.505	840.654	12	13	7
9	190	207	230.581	316.936	5	4	3
10	201	248	257.936	351.706	6	5	2
11	333	502	422.877	717.914	9	11	9
12	325	657	450.463	964.21	8	11	13
13	159	520	278.522	676.872	4	8	8
14	21	21	35.1421	35.1421	3	1	0
15	222	241	251.723	272.794	5	3	3
16	348	761	425.392	928.938	10	8	13


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 16)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 16813)
count
width (px)